



Coverage Metrics for T-Wise Feature Interactions

SE'2026 | Sabrina Böhm, Tim Schmidt, Sebastian Krieter, Tobias Pett, Thomas Thüm, Malte Lochau | February 25, 2026 | Bern



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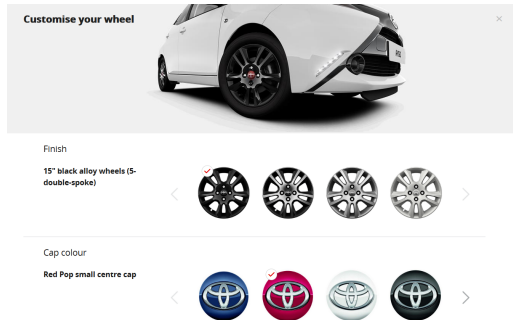


Technische
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Braunschweig

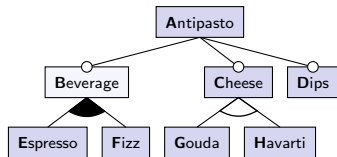


Software Engineering
Programming Languages

Configurable Systems

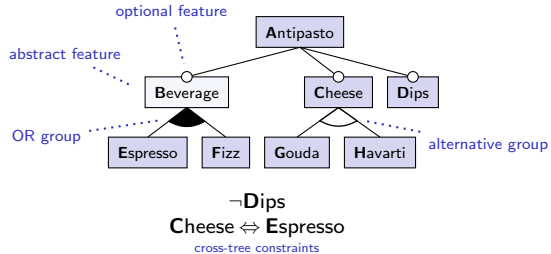


Modeling the Configuration Space

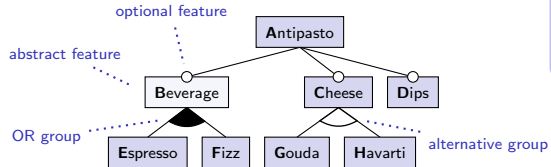


$\neg \text{Dips}$
 $\text{Cheese} \Leftrightarrow \text{Espresso}$

Modeling the Configuration Space



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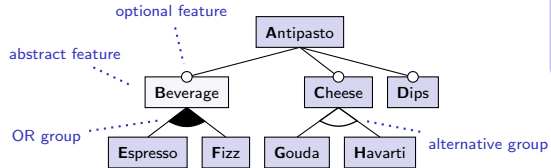


$\neg \text{Dips}$
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cross-tree constraints

Valid Configuration

$$c_1 = \{A, \neg B, \neg C, \neg D, \neg E, \neg F, \neg G, \neg H\}$$

Modeling the Configuration Space



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cross-tree constraints

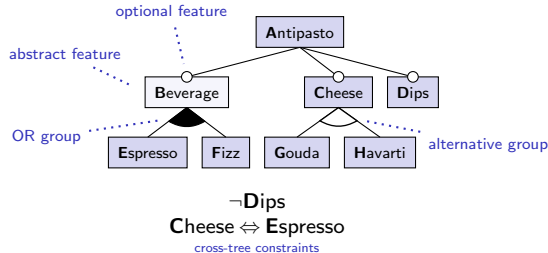
Valid Configuration

$c_1 = \{A, \neg B, \neg C, \neg D, \neg E, \neg F, \neg G, \neg H\}$

$c_2 = \{A, B, C, \neg D, E, \neg F, G, \neg H\}$

...

Modeling the Configuration Space



Valid Configuration

$c_1 = \{A, \neg B, \neg C, \neg D, \neg E, \neg F, \neg G, \neg H\}$

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...

Product-Based Testing

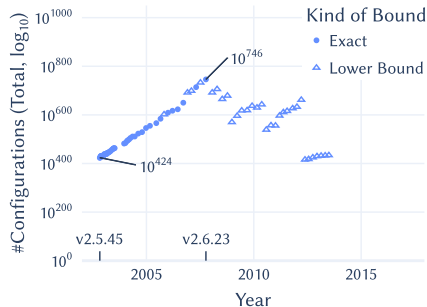
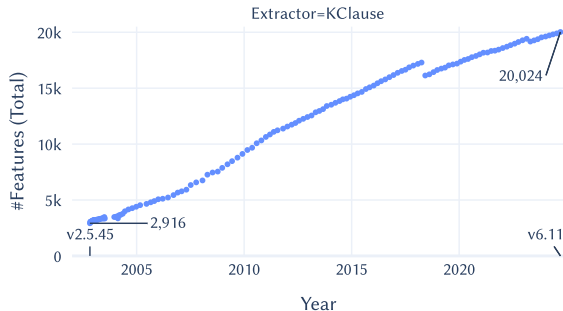
Run test suite for individual configurations.

$\text{test}(\text{build}(c_1))$

$\text{test}(\text{build}(c_2))$

...

Configurable Software Systems – The Linux Kernel



⇒ Testing every configurations is infeasible!

T-Wise Interaction Testing

T-Wise Interaction Sampling

Motivation: Find small, yet representative list of configurations

Idea: Generate sample such that **every valid interaction** between t features is contained in **at least one configuration**.

T-Wise Interaction Testing

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Partial Sample

Samples that do not achieve 100% coverage for a given criterion

- + Even smaller
- + Faster to generate
- May miss some crucial interaction

T-Wise Interaction Testing

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Partial Sample

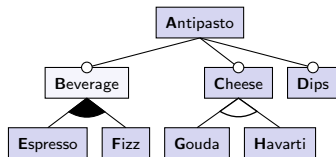
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- + Even smaller
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Partial Sample Coverage

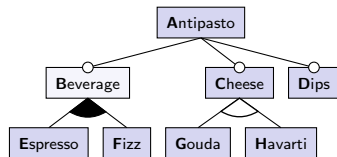
- What coverage percentage is achieved?
- Which interactions are missing?

Pair-Wise Interaction Coverage



$\neg \text{Dips}$
 $\text{Cheese} \Leftrightarrow \text{Espresso}$

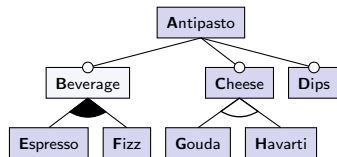
Pair-Wise Interaction Coverage



$\neg \text{Dips}$
 $\text{Cheese} \Leftrightarrow \text{Espresso}$

	A	$\neg A$	B	$\neg B$	C	$\neg C$	D	$\neg D$	E	$\neg E$	F	$\neg F$	G	$\neg G$
$\neg H$														
H														
$\neg G$														
G														
$\neg F$														
F														
$\neg E$														
E														
$\neg D$														
D														
$\neg C$														
C														
$\neg B$														
B														

Pair-Wise Interaction Coverage



$\neg$ Dips
Cheese $\Leftrightarrow$ Espresso

Sample

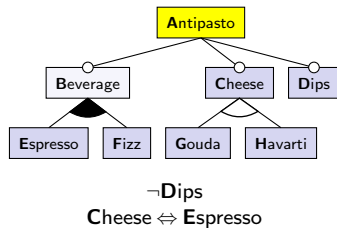
$S = \{ \{A, \neg B, \neg C, \neg D, \neg E, \neg F, \neg G, \neg H\}, \\ \{A, B, C, \neg D, E, \neg F, G, \neg H\} \}$

	A	$\neg A$	B	$\neg B$	C	$\neg C$	D	$\neg D$	E	$\neg E$	F	$\neg F$	G	$\neg G$
$\neg H$	✓		✓	✓	✓	✓		✓	✓		✓	✓	✓	✓
H												✓		
$\neg G$	✓			✓		✓		✓		✓		✓		
G	✓		✓		✓			✓	✓			✓		
$\neg F$	✓		✓	✓	✓	✓		✓	✓	✓				
F														
$\neg E$	✓			✓		✓		✓						
E	✓		✓		✓			✓						
$\neg D$	✓		✓	✓	✓	✓								
D														
$\neg C$	✓				✓									
C	✓		✓											
$\neg B$	✓													
B	✓													

Pair-Wise Coverage

$\text{cov}(S) \approx 69\%$

Pair-Wise Coverage – Without Core Features



Sample

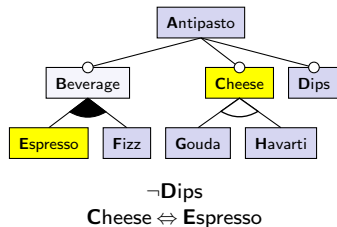
$$S = \{ \{A, \neg B, \neg C, \neg D, \neg E, \neg F, \neg G, \neg H\}, \\ \{A, B, C, \neg D, E, \neg F, G, \neg H\} \}$$

	A	¬A	B	¬B	C	¬C	D	¬D	E	¬E	F	¬F	G	¬G
¬H			✓	✓	✓	✓		✓	✓		✓	✓	✓	✓
H												✓		
¬G				✓		✓		✓		✓		✓		
G			✓		✓			✓	✓			✓		
¬F			✓	✓	✓	✓		✓	✓	✓				
F														
¬E				✓		✓		✓						
E			✓		✓			✓						
¬D			✓	✓	✓	✓								
D														
¬C					✓									
C			✓											
¬B														
B														

Pair-Wise Coverage

$$\text{cov}(S) \approx 66\%$$

Pair-Wise Coverage – Without Atomic Feature Sets



Sample

$$S = \{ \{A, \neg B, \neg C, \neg D, \neg E, \neg F, \neg G, \neg H\}, \\ \{A, B, C, \neg D, E, \neg F, G, \neg H\} \}$$

	A	¬A	B	¬B	C	¬C	D	¬D	E	¬E	F	¬F	G	¬G
¬H	✓		✓	✓	✓	✓		✓			✓	✓	✓	✓
H					✓							✓		
¬G	✓			✓	✓	✓		✓				✓		
G	✓		✓		✓			✓				✓		
¬F	✓		✓	✓	✓	✓		✓						
F					✓	✓								
¬E														
E														
¬D	✓		✓	✓	✓	✓								
D														
¬C	✓				✓									
C	✓		✓											
¬B	✓													
B	✓													

Pair-Wise Coverage

$$\text{cov}(S) \approx 70\%$$

General Coverage Calculation

Default Coverage

$$\diamond(M, t) \mapsto \{i \subseteq L(F) : |i| = t, \exists c \in C(M) : i \subseteq c\}$$

$$\text{cov}(t, M, S) \mapsto \begin{cases} 1 & , \text{ if } |\diamond(M, t)| = 0 \\ \frac{|\{i \in \diamond(M, t) : \exists c \in S : i \subseteq c\}|}{|\diamond(M, t)|} & , \text{ else} \end{cases}$$

General Coverage Calculation

Filtered Coverage

$$\text{⬠}(M, t) \mapsto \{i \subseteq L(F) : |i| = t, \exists c \in C(M) : i \subseteq c\}$$

$$\text{cov}(\text{⬢}, t, M, S) \mapsto \begin{cases} 1 & , \text{ if } |\text{⬢}(\text{⬠}(M, t))| = 0 \\ \frac{|\{i \in \text{⬢}(\text{⬠}(M, t)) : \exists c \in S : i \in c\}|}{|\text{⬢}(\text{⬠}(M, t))|} & , \text{ else} \end{cases}$$

General Coverage Calculation

Filtered Coverage

$$\boxtimes(M, t) \mapsto \{i \subseteq L(F) : |i| = t, \exists c \in C(M) : i \subseteq c\}$$

$$\text{cov}(\Upsilon, t, M, S) \mapsto \begin{cases} 1 & , \text{ if } |\Upsilon(\boxtimes(M, t))| = 0 \\ \frac{|\{i \in \Upsilon(\boxtimes(M, t)) : \exists c \in S : i \subseteq c\}|}{|\Upsilon(\boxtimes(M, t))|} & , \text{ else} \end{cases}$$

$$\Upsilon_{\text{default}}(I) = I$$

$$\Upsilon_{CF}(I) = \{i \in I : i \cap \text{core}(M) = \emptyset\}$$

$$\Upsilon_{DF}(I) = \{i \in I : i \cap \text{dead}(M) = \emptyset\}$$

$$\Upsilon_{AF}(I) = \{i \in I : i \cap \text{abstract}(M) = \emptyset\}$$

$$\Upsilon_{ArbF}(I) = \{i \in I : i \cap I_{\text{filter}} = \emptyset\}$$

General Coverage Calculation

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$$\Upsilon_{AFS}(I) = \{i \in I : \forall a \in \text{atomicFeatureSets}(M) : |a| > 1 \Rightarrow i \cap (a \setminus \{\neg \Pi_1(a), \Pi_1(a)\}) = \emptyset\}$$

$$\Upsilon_{ALS}(I) = \{i \in I : \forall a \in \text{atomicLiteralSets}(M) : |a| > 1 \Rightarrow i \cap (a \setminus \{\Pi_1(a)\}) = \emptyset\}$$

$$\Upsilon_{PCI}(I) = \{i \in I : \forall pc \in \text{parentChild}(M) : pci \not\subseteq i\}$$

$$\Upsilon_{ArbI}(I) = I \setminus I_{\text{filter}}$$

Coverage Metrics in Literature

✓ filter is applied

⊘ filter is not applied

? unsure

Metric	CF	DF	AF	ArbF	AFS	ALS	PCI	ArbI	#Paper
M1	✓	✓	⊘	✓	✓	⊘	⊘	⊘	1
M2	✓	✓	⊘	✓	⊘	⊘	⊘	✓	2
M3	⊘	⊘	✓	✓	⊘	⊘	⊘	⊘	1
M4	✓	✓	⊘	⊘	⊘	⊘	⊘	⊘	5
M5	⊘	⊘	⊘	⊘	⊘	⊘	✓	⊘	1
M6	⊘	⊘	⊘	⊘	⊘	⊘	⊘	⊘	6
	?/⊘	?/⊘	?/⊘	?/⊘	?/⊘	?/⊘	?/⊘	?/⊘	30

Evaluation Setup

Research Questions

RQ_1 Does the choice of a **metric impact** the **coverage value**?

RQ_2 How does the choice of a **metric** and **number of features** affect the **number of filtered interactions**?

RQ_3 How does the choice of a **metric** and **number of features** affect the **computation time**?

Evaluation Setup

Research Questions

RQ₁ Does the choice of a **metric impact** the **coverage value**?

RQ₂ How does the choice of a **metric** and **number of features** affect the **number of filtered interactions**?

RQ₃ How does the choice of a **metric** and **number of features** affect the **computation time**?

Feature Models:

- 48 toy and realistic feature models
- Domains: finance, systems software, e-commerce, gaming, and communication
- Number of features:
9 – 2,513
- Number of constraints:
13 – 15,692

Evaluation Setup

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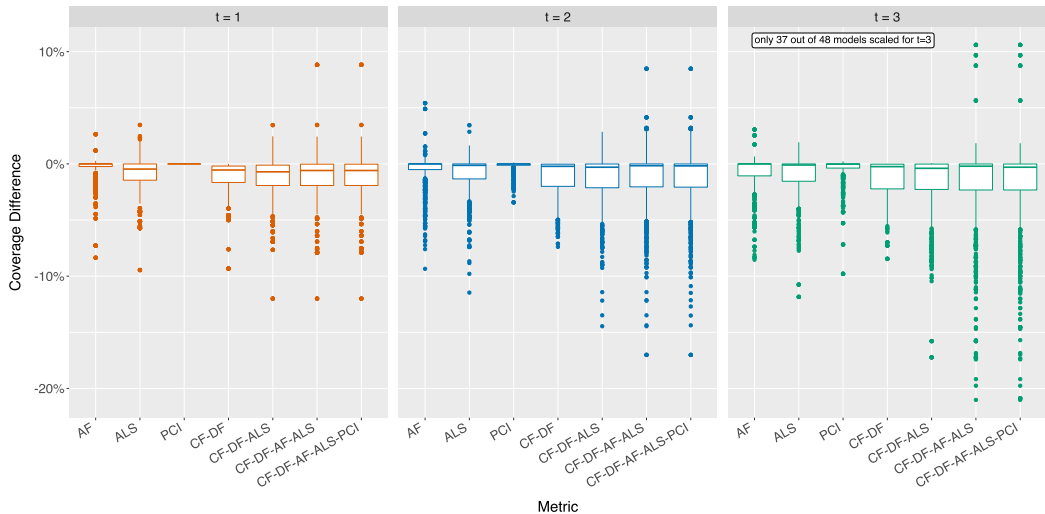
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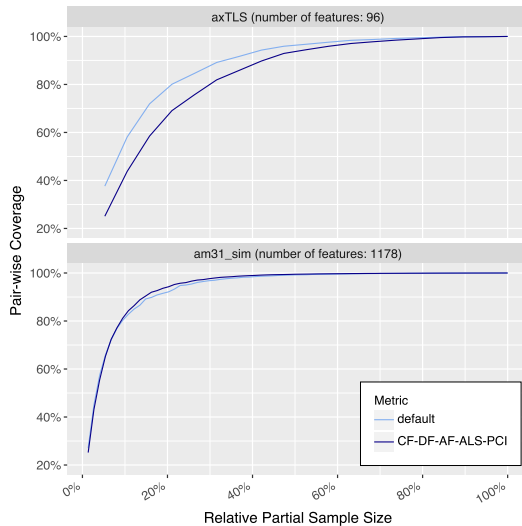
Selected Filter Combinations:

- default filter (i.e., $\mathbf{T}_{default}$, called default metric) (**M6**)
- abstract filter (i.e., $\mathbf{T}_{AF}$) (**M3** omitting arbitrary features)
- atomic literal set filter (i.e., $\mathbf{T}_{ALS}$)
- parent child interactions filter (i.e., $\mathbf{T}_{PCI}$) (**M5**)
- core and dead filters (i.e., $\mathbf{T}_{CF-DF} = \mathbf{T}_{CF} \circ \mathbf{T}_{DF}$) (**M4** and **M2** without arbitrary features and interactions)
- filters core, dead, and atomic literal set $\mathbf{T}_{CF-DF-ALS} = \mathbf{T}_{CF} \circ \mathbf{T}_{DF} \circ \mathbf{T}_{ALS}$ (**M1** without arbitrary features)
- filters core, dead, atomic literal set, abstract, and parent child interactions $\mathbf{T}_{CF-DF-AF-ALS-PCI} = \mathbf{T}_{CF} \circ \mathbf{T}_{DF} \circ \mathbf{T}_{AF} \circ \mathbf{T}_{ALS} \circ \mathbf{T}_{PCI}$

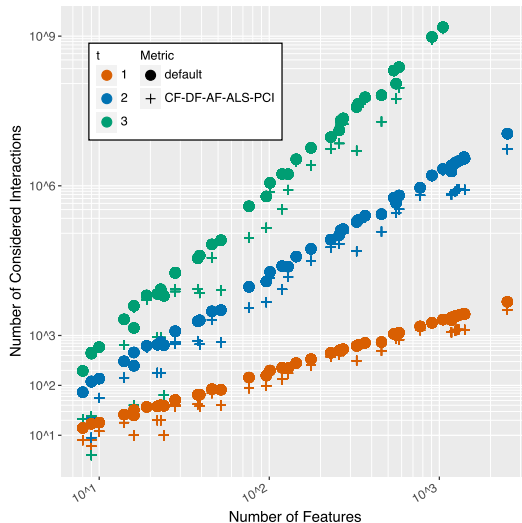
RQ₁: Coverage Value per Metric



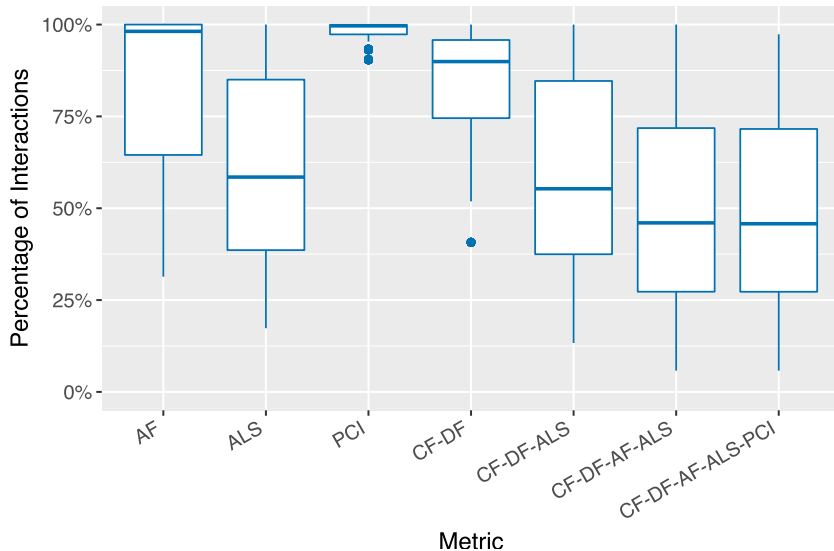
RQ₁: Coverage Value per Metric – Example



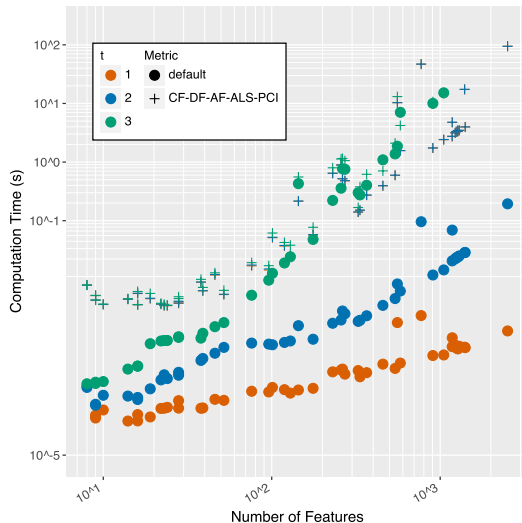
RQ_{2a}: Number of Interactions per Number of Features



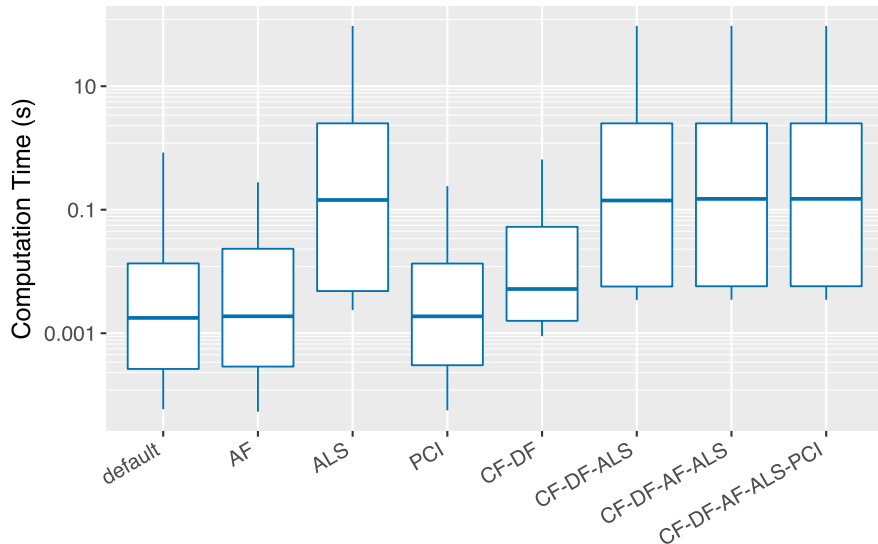
RQ_{2b}: Number of Interactions per Metric



RQ_{3a}: Computation Time per Number of Features



RQ_{3b}: Computation Time per Metric



Conclusion

- No consensus on t -wise feature interaction coverage in the literature
 - ⇒ No comparability across the literature.
 - ⇒ Coverage values may differ up to 21%
- ⇒ First systematic framework on coverage metrics
 - New studies can report on used metric
 - We recommend to exclude core and dead features



Metric	CF	DF	AF	ArbF	AFS	ALS	PCI	Arbl	#Paper
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M5	⊘	⊘	⊘	⊘	⊘	⊘	✓	⊘	1
M6	⊘	⊘	⊘	⊘	⊘	⊘	⊘	⊘	6
	?/⊘	?/⊘	?/⊘	?/⊘	?/⊘	?/⊘	?/⊘	?/⊘	30

